

Employee Turnover Prediction

A machine learning approach to identifying at-risk employees and enabling proactive retention strategies before valuable talent walks out the door.

HR ANALYTICS PROJECT OVERVIEW



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Why Turnover Prediction Matters

Voluntary resignation — unlike unplanned attrition — is both predictable and preventable. The organizational cost is substantial and multi-dimensional.

Recruitment Costs

Replacing a single employee can cost 50–200% of their annual salary in sourcing, interviewing, and onboarding.

Institutional Knowledge

Departing employees take with them years of domain expertise, relationships, and organizational context.

Team Continuity

High turnover disrupts project momentum, strains remaining team members, and reduces overall productivity.



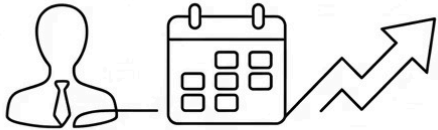
Dataset & Feature Overview

Source Files

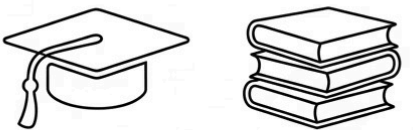
Two datasets power this project:

- **hr_analytics.csv** — Raw HR dataset with original employee records
- **data.csv** — Processed and encoded version used directly for modeling

Together, these files provide a clean pipeline from raw HR data to model-ready input.



Role and Tenure:
Position, Years at
Company, Years
Since Last
Promotion



Education and
Experience:
Education Level,
Total Working
Years



Wellbeing:
Relationship
Satisfaction, Work-
Balance, Number of
Companies Worked



Personal and
Financial:
Marital Status, Age,
Monthly Income,
Stock Option Level



Project Structure

The codebase is organized for modularity, reproducibility, and deployment-readiness.

1

data/

Raw and processed datasets stored separately for traceability.

2

notebooks/

Six Jupyter notebooks covering EDA, preprocessing, modeling, evaluation, and interpretation.

3

functions/

Modular `.py` utility files separated for readability and reuse across notebooks.

4

requirements/

Serialized best model in `.joblib` format, ready for Streamlit deployment.

Preprocessing Strategy

What Was Done

- Categorical variables encoded for model compatibility
- Dataset arrived in good condition — minimal cleaning required
- Natural class distribution preserved intentionally

Key Design Decision

The dataset exhibits significant **class imbalance** — retained employees far outnumber those who left. Rather than correcting this through oversampling or undersampling, the natural distribution was preserved to reflect real-world conditions and addressed through model selection instead.



Exploratory Data Analysis

EDA combined summary statistics with visualizations to surface patterns across all feature domains — before any modeling took place.

→ Distribution Analysis

Feature distributions were examined to identify skewness, outliers, and relationships between variables.

→ Class Imbalance Confirmed

Visualizations confirmed that retained employees substantially outnumber departures — reinforcing the need for a model suited to imbalanced classification.

→ Retention Signal Identified

Several features showed strong separation between retained and departed employees, validating their inclusion in the final model.



Model Selection: Random Forest Classifier

Why Random Forest?

- Robust to class imbalance without requiring resampling
- Captures non-linear relationships between HR features
- Ensemble approach reduces overfitting risk
- Provides native feature importance rankings

Hyperparameter Tuning

GridSearchCV was used to systematically identify the optimal model configuration prior to final fitting, ensuring the best possible performance on held-out data.

Accuracy

0.85 — strong overall performance, especially alongside precision and recall metrics.

Evaluation

Full per-class classification report available in `notebooks/3_model_selection`.

Model Performance at a Glance

0.85

Accuracy

Overall classification accuracy on the held-out test set.

4

Feature Domains

Predictive signals spanning role, education, wellbeing, and personal factors.

15+

Input Features

Distinct HR attributes used to score each employee's departure risk.

The Random Forest model delivers reliable risk scores across both the majority (retained) and minority (departed) classes — a critical bar for real-world deployment in HR settings.

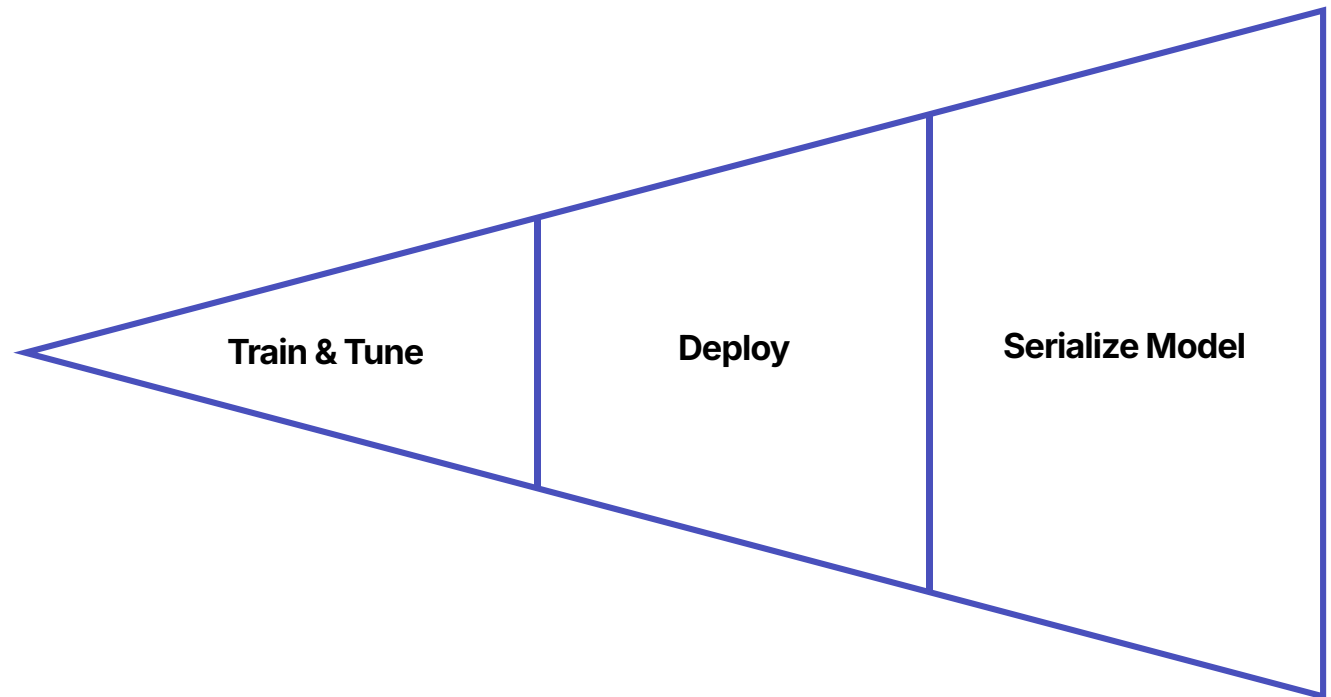
Model Interpretation & Storage

Interpretation

Detailed performance analysis — including **feature importance rankings** and per-class precision/recall — is documented in `notebooks/3_model_selection`. Feature importance enables HR teams to understand *which factors most strongly predict departure risk*, supporting explainable AI practices.

Model Storage & Deployment

The best-performing model is serialized using `joblib` and stored in the `requirements/` directory. A dedicated **model_implementation notebook** supports loading and testing on unseen data. The model is ready for integration into a **Streamlit application** for interactive inference.



Key Takeaways & Next Steps

Turnover is Predictable

Machine learning can surface at-risk employees early — creating a window for targeted retention intervention before departure decisions are made.

Real-World Design Choices

Preserving class imbalance and selecting a robust algorithm (Random Forest) ensures the model generalizes to real HR populations — not just balanced test sets.

Deployment-Ready

With the model serialized and a Streamlit implementation underway, the next step is integrating live HR data feeds for continuous, automated risk scoring.

 Reference: S. Flowers (1974), *Why Employees Stay* — Harvard Business Review. Dataset via Kaggle: Employee Attrition and Factors.

